1. Goal

The goal of this project is to build an interactive product about (choose one):

1. Extreme Programming and its integration with Scrum
2. Kanban and flow metrics (lead time, throughput, cumulative flow)
3. DevOps and Platform Engineering (incl. DORA metrics)
4. MLOps: testing, CI/CD and monitoring of Machine Learning systems
5. Lean Startup and product discovery for Data/AI products
6. Scrum in Education
7. Value Stream Mapping in Agile Projects
8. Agile Coach
9. Agile in Non-Technical Projects
10. Gamifying Agile Practices for Team Engagement
11. Agile Practices for Hybrid and Distributed Teams
12. Management 3.0
13. Agile development of LLM and chatbot products (evaluation-driven development)
14. Human–AI teaming: Scrum roles, events and Definition of Done when AI agents write code
15. Shape Up and other alternatives to Scrum
16. (other – please propose)

This product shall be built in an agile way. This way, students shall acquire an experience on managing an agile project, using practices related with, but not limited to, **Scrum**. The final goal is to present both the interactive product, and a report about the agile experience.

2. Product

The product to develop shall be an **interactive webpage**. The product shall run on Netlify. It will also include a chatbot that answers user questions related to the webpage using BOTSchool.ai. Additionally, the students are incentivized to also use Lovable.dev to brainstorm ideas and mockups.

The code of the webpage can be any kind of Javascript library. The tools to manage the project are also the team's choice. This interactive product should engage someone to learn about the subject. It should be both informative and interactive. It should be visually attractive. The user may be able to interact with the presentation, by following different paths, answering quizzes, etc.
The time to develop the product (write, review, design, etc.) will be of:

- **P1 (Tuesday class):** 1 Design Thinking session + 2 sprints x 3 "days" per sprint x 2 hours per day x N persons per team (e.g. in a 5 persons team, 60 hours + the Design Thinking session).

- **P2 (Monday class):** 1 Design Thinking session + 3 sprints x 3 "days" per sprint x 2 hours per day x N persons per team (e.g. in a 5 persons team, 90 hours + the Design Thinking session).

Planning and research regarding the theme is not time-bounded and not controlled. This product shall be demonstrated at the end of each sprint.
The scope of the product is assessed relative to the time available to each class (P1: 2 sprints; P2: 3 sprints).

Use of AI tools. AI tools (e.g. Lovable.dev, AI coding agents, LLM chatbots) may be used, but their use must be declared in the report: which tools, and for what. The team remains fully responsible for the correctness of all content and code.

Chatbot. The BOTSchool.ai knowledge base should be built from the sources referenced by the team. The team should keep a small test set of questions with expected answers and use it to check the chatbot at each sprint.

Definition of Done. Each team defines its Definition of Done at Release Planning. It should include, at least: responsive and accessible webpage; at least one quiz or interactive path; deployed on Netlify; chatbot answers the test set correctly.

3. Delivers, Report and final demo

There is a single delivery, at the end of the project (see section 6). Intermediate artifacts (backlogs, burndown charts, meeting minutes, etc.) must be kept up to date in the team's tools during the project and included in the final delivery.

There will be a report and a presentation, describing the experience within the project. The group might include extra documents in pdf or doc format to clarify some points. The style, size, and contents are free of choice. The report should include:

- Design Thinking result and output.
- Project lifecycle
- Practices, roles, artifacts, charts.
- Meeting minutes.
- Analysis (team and personal)
- Metrics used (e.g. velocity, burndown) and their interpretation.
- Comparison between the Design Thinking output and the final product.
- AI usage statement (which tools were used, and for what).

The time to develop the report is not to be controlled – it will be as much as necessary. The presentation shall take no longer than 15 minutes, followed by 5 minutes discussion. The teams should support very well their choices. Reading and referencing external bibliography is mandatory and well appreciated.

4. Rules

- The project will have one release, and 2 (P1) or 3 (P2) sprints. Each sprint has 3 working "days" of 2 hours. This will be during and out of class work.
- An online tool shall be used for the story cards / requirements, product backlog, sprint backlog, burndown chart, among others.
- The team must demo something at the end of each sprint.
- The team must conduct their daily scrum meetings and update the burndown chart "daily".
- The Scrum Master is part of the team.
- The Product Owner is part of the team.
- The teacher plays the role of the Client.
- Use as many agile rules and practices as possible.
- Time and schedule are presented in next section.
- The report should be written along the project development. This time is not recorded. Take evidence of the process (e.g. screenshots, photos, etc.).
- Each team must choose its members and a theme by **27/October (P1, Tuesday class: at the start of the class) or 02/November (P2, Monday class)**, by writing in the **Wiki (P1 or P2)** the theme, a short description, and the names of the team members
- The team should have between 4 to 5 people.
- Each team must define its Definition of Done at Release Planning (see Section 2).

5. Assessment

Team grade. The team grade considers the agile process (practices, artifacts and evidence), the report, the product and the presentation. The agile process and the quality of the critical analysis are valued above the polish of the product.

Teacher will assess the teams, not the individual students.
Then, each student will assess each one of his colleagues, according to the statements below, and then obtain a relative assessment of each team member. The team should then meet and distribute the team grade among the team members, assuming that the team grade is the average of the grades of its members.
The statements that will be used for assessing each team member, using a scale 1-4 (1=strongly disagree; 2=disagree; 3=agree; 4=strongly agree), are the following:

1. Attend group meetings regularly and arrives on time.

2. Contributes meaningfully to group discussions.
3. Completes group assignments on time.
4. Prepares work in a quality manner.
5. Demonstrates a cooperative and supportive attitude.
6. Contributes significantly to the success of the project

6. Working time and schedule

P1 – Tuesday class (2 sprints)

- **27/October** (during class)
    - **Sprint 0/Pre-game – Design Thinking (60 min):** understand the client needs and possible solutions to the problem.
    - **Release Planning (30 min):** review user stories, estimate, create a release plan, define the Definition of Done.
    - **Sprint 1 Planning Meeting (30 min):** sprint goal + planning.
- **Sprint 0/Pre-game** (out of class, no time-limit)
    - Refine project goals, vision, general requirements and roles; choose tools (backlog, burndown chart format, etc.); write and refine user stories. Ask the instructor for clarifications if necessary.
- **Week 27/Oct – 10/Nov:** Sprint 1 (3 x 2 hours) – the first "day" during class on 03/November
- **10/November** (during class)
    - Sprint 1 Review and Demo (15 min presentation each team)
    - Sprint 1 Retrospective (10 min)
    - **Sprint 2 Planning Meeting (30 min):** sprint goal + planning.
- **Week 10/Nov – 17/Nov:** Sprint 2 (3 x 2 hours) – the first "day" during class on 10/November
- **17/November** (during class)
    - Sprint 2 Review and Demo (15 min presentation each team)
    - Sprint 2 Retrospective (10 min)
    - Prepare final presentation
    - Final Presentation (15 min + 5 min discussion each team)
        - **Final deliverables:** Design Thinking output; Project plan (goals, vision, general requirements, release plan, rules, Definition of Done); Product backlog; Sprint goals, sprint backlogs and burndown charts; Review and retrospective meeting minutes; Product (Netlify URL); Final report; Presentation; Link to the backlog tool.

P2 – Monday class (3 sprints)

- **02/November** (during class)

    - **Sprint 0/Pre-game – Design Thinking:** understand the client needs and possible solutions to the problem.
- **Sprint 0/Pre-game** (out of class, no time-limit)
    - Define project goals, vision, general requirements and roles; choose tools (backlog, burndown chart format, etc.); write user stories. Ask the instructor for clarifications if necessary.
- **09/November** (during class)
    - **Release Planning (45 min):** review user stories, estimate, create a release plan, define the Definition of Done.
    - **Sprint 1 Planning Meeting (45 min):** sprint goal + planning.
- **Week 09/Nov – 16/Nov:** Sprint 1 (3 x 2 hours) – the first "day" during class on 09/November
- **16/November** (during class)
    - Sprint 1 Review and Demo (15 min presentation each team)
    - Sprint 1 Retrospective (10 min)
    - **Sprint 2 Planning Meeting (30 min):** sprint goal + planning.
- **Week 16/Nov – 23/Nov:** Sprint 2 (3 x 2 hours) – the first "day" during class on 16/November
- **23/November** (during class)
    - Sprint 2 Review and Demo (15 min presentation each team)
    - Sprint 2 Retrospective (10 min)
    - **Sprint 3 Planning Meeting (30 min):** sprint goal + planning.
- **Week 23/Nov – 30/Nov:** Sprint 3 (3 x 2 hours) – the first "day" during class on 23/November
- **30/November** (during class)
    - Sprint 3 Review and Demo (15 min presentation each team)
    - Sprint 3 Retrospective (10 min)
    - Prepare final presentation (45 min)
    - Final Presentation (15 min + 5 min discussion each team)
        - **Final deliverables:** Design Thinking output; Project plan (goals, vision, general requirements, release plan, rules, Definition of Done); Product backlog; Sprint goals, sprint backlogs and burndown charts; Review and retrospective meeting minutes; Product (Netlify URL); Final report; Presentation; Link to the backlog tool.

7. Example Tools

- Jira – https://www.atlassian.com/software/jira
- Trello – https://trello.com/
- GitHub Projects – https://docs.github.com/issues/planning-and-tracking-with-projects
- Azure Boards – https://azure.microsoft.com/products/devops/boards
- Linear – https://linear.app/
- Taiga – https://taiga.io/

- Miro (Design Thinking, retrospectives) – https://miro.com/

8. References

1. Larman, Craig. Agile and Iterative Development: A Manager's Guide. 2004.(Chapter 7 & 8)
2. Schwaber, Ken. Agile Project Management with Scrum, 2004.
3. Extreme Programming http://www.extremeprogramming.org/
4. Agile Software Development http://www.agile-process.org/
5. Scrum and XP from the Trenches http://www.infoq.com/minibooks/scrum-xp-from-the-trenches-2
6. Rostal, Pam. 59-Minute Scrum. http://www.otug.org/meeting/slides/20060919-59_Minute_Scrum.ppt
7. http://www.scrumalliance.org
8. http://www.mountaingoatsoftware.com/
9. Andre Stellman & Jennifer Greene. Learning Agile
10. Schwaber, Ken & Sutherland, Jeff. The Scrum Guide, 2020. https://scrumguides.org
11. Vacanti, Daniel & Yeret, Yuval. The Kanban Guide, 2020. https://kanbanguides.org
12. Forsgren, Nicole, Humble, Jez & Kim, Gene. Accelerate: The Science of Lean Software and DevOps. IT Revolution, 2018.
13. Kreuzberger, Dominik, Kühl, Niklas & Hirschl, Sebastian. Machine Learning Operations (MLOps): Overview, Definition, and Architecture. IEEE Access, 2023.
14. Ries, Eric. The Lean Startup. Crown Business, 2011.
15. Singer, Ryan. Shape Up: Stop Running in Circles and Ship Work that Matters. Basecamp, 2019. https://basecamp.com/shapeup
16. Rother, Mike & Shook, John. Learning to See: Value Stream Mapping to Add Value and Eliminate Muda. Lean Enterprise Institute, 1999.
17. Adkins, Lyssa. Coaching Agile Teams. Addison-Wesley, 2010.
18. Appelo, Jurgen. Management 3.0: Leading Agile Developers, Developing Agile Leaders. Addison-Wesley, 2011.